PASS SCIENCE WORKS | R&D INTERNSHIP 2026–2027

Unified Engineering Workbench — Engineering Report

Prototype status: C++17 numerical/data core implemented and locally verified. The Windows one-click GUI and interactive 3D visualizations are not yet complete; see limitations.

1. Design and architecture

The prototype separates numerical/data functions in `src/workbench.cpp` from build configuration and automated tests. The C++ executable exposes thermal FEA, deterministic batch benchmarking, STL/CSV validation, hardware mocks and backpropagation. CMake provides a reproducible source build. A production submission should add a presentation layer over these APIs while keeping numerical and file-processing logic in C++.

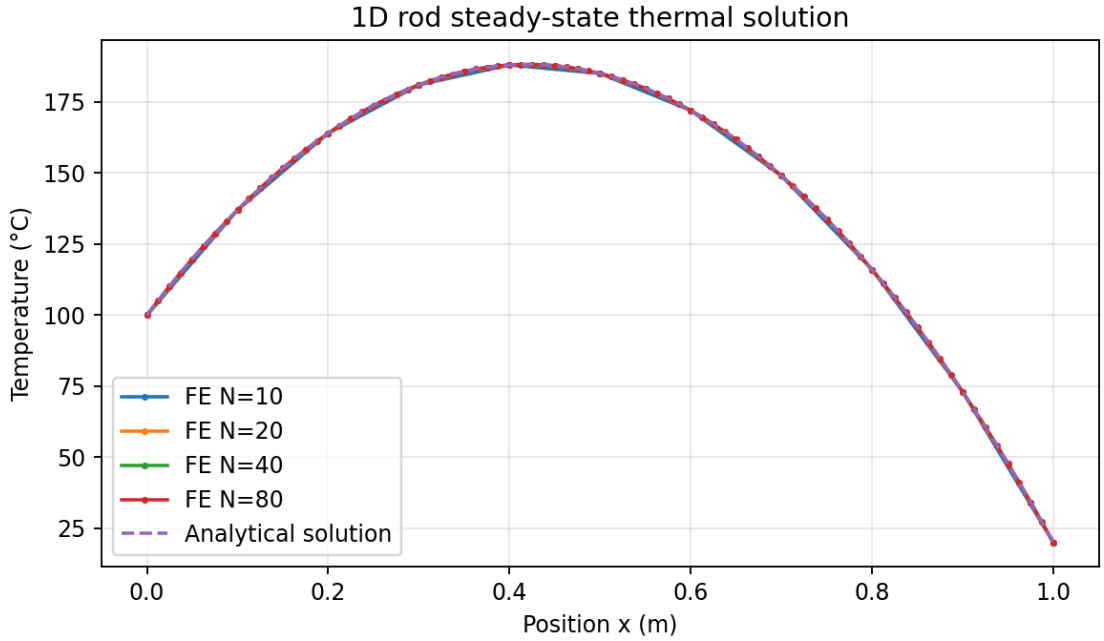
2. Thermal FEA formulation and verification

Solve `-d/dx(k A dT/dx) = q A` on a 1 m rod with A=1e-4 m², k=10 W/(m K), q=10,000 W/m³, T(0)=100 °C, T(L)=20 °C and insulated lateral surfaces. Two-node linear elements use Ke=(kA/le)[[1,-1],[-1,1]] and fe=(qA le/2)[1,1]■. Prescribed endpoint temperatures are imposed by eliminating endpoint degrees of freedom.

The analytical solution is T_exact(x)=100–80x/L+[q/(2k)]x(L–x). Midpoint error is evaluated using FE interpolation. For this quadratic solution, error=q le²/(8k)=125/N² °C.

Elements N	Max midpoint error (°C)
10	1.250000
20	0.312500
40	0.078125
80	0.019531

Error decreases by four whenever N doubles, consistent with second-order convergence. Debugging insight: the quadratic coefficient is q/(2k)=500, not q/k; missing the factor of two doubles the midpoint error.



3. CAD / nodal result imports

The STL parser accepts ASCII facet/vertex records and rejects incomplete, empty or non-finite triangle data. The example is a self-created unit cube with 12 triangles. The CSV parser requires exactly node_id,x_m,y_m,z_m,temperature_C, validates finite numeric values and rejects empty datasets. The included CSV is explicitly synthetic, not an actual ANSYS export. Coordinates are metres and temperature is Celsius. Current prototype validates imports but does not yet render interactive 3D views, a temperature legend or node inspection.

4. Multicore execution

The deterministic generator uses seed 20261101, 2,000 independent cases, 80 elements per case, k in [5,25] W/(m K) and q in [1,000,20,000] W/m³. Serial and parallel paths use the same solver. An atomic work index assigns cases to a bounded std::thread pool; output order remains deterministic. One warm-up is excluded, five measured repetitions are used and medians are reported. I/O and rendering are excluded.

Workers	Median s	Speedup	Efficiency	Max ΔT °C
1	0.218792	1.000	1.000	0
2	0.113629	1.926	0.963	0
3	0.076636	2.855	0.952	0
4	0.063874	3.425	0.856	0

Measured host: AMD EPYC 9V74 80-Core Processor as exposed to this runtime; one socket, five available logical CPUs/five exposed cores, 5.8 GiB RAM. Four-worker efficiency is below 1 due to scheduling/thread overhead. These container measurements are not representative of the candidate Windows machine; repeat there and record launch-to-ready time.

5. Embedded integration design

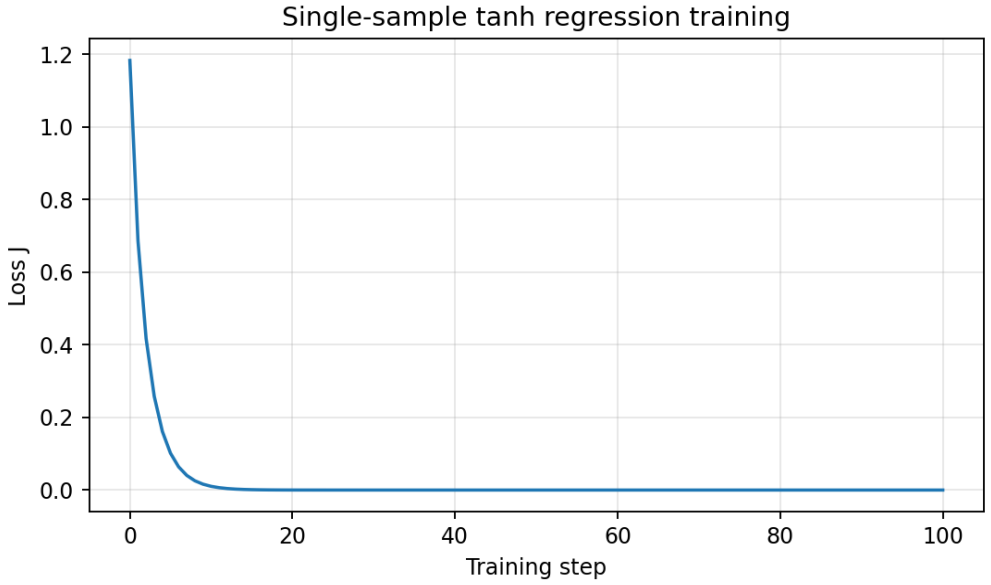
The accompanying embedded-design document includes a host/MCU/FPGA/JTAG block diagram and interface/pin-voltage assumptions. C++ mocks implement discovery, capability reporting, connect/disconnect and simulated telemetry. A forced FPGA timeout is reported as recoverable. A JTAG adapter is a debug/programming transport, not an interchangeable compute board. Verify exact board manuals, target Vref, pinout and connector before any physical wiring. No hardware was connected.

6. Handwritten backpropagation

For x=1, y=2, h=tanh(wx+b), y_hat=vh+c and J=0.5(y_hat–y)², the initial forward pass gives h=0.4621171573, y_hat=0.4621171573 and J=1.182541819. With e=y_hat–y, the gradients are dJ/dc=e, dJ/dv=e h, dJ/dw=e v(1–h²)x and dJ/db=e v(1–h²).

Quantity	Value
dw	–1.209464475240
db	–1.209464475240
dv	–0.710682047486
dc	–1.537882842740
w_new	0.620946447524
b_new	0.120946447524
v_new	1.071068204748
c_new	0.153788284274

All gradients are negative, so subtracting 0.1 times each gradient increases all four parameters. Central finite differences with epsilon=1e-6 agree with analytic gradients to much better than 1e-5.



7. Test outcomes and limitations

Local CMake build and CTest suite pass. Tests cover FEA midpoint errors and boundary values, serial/parallel agreement within 1e-8 °C, invalid mesh rejection and analytic-vs-finite-difference gradients. Example STL and CSV files pass import validation.

Outstanding assignment-critical work: responsive UI with progress/cancellation/export; interactive STL and 3D temperature point-cloud views; node inspection and legend; Windows 11 Launch.exe packaging and clean-machine validation; editor one-action run configuration for the packaged app; target-machine metadata and launch-to-ready timing; timestamped demo evidence; and the required handwritten (maximum two-page) appendix. No Windows launcher or real hardware integration is included in this prototype.